

CANDIDATE NAME _____

INDEX NUMBER _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 Preliminary Examination 2016

H2 BIOLOGY
9648

Paper 2

2 Hours

Additional materials:
Writing papers

Date / Day: 13th September 2016/Tuesday**INSTRUCTIONS TO CANDIDATES**

Write your name, CG and index number in the spaces at the top of this page and on all separate writing papers used.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Section A

Answer **all** questions.

Write your answers in spaces provided on the question paper.

Section B

Answer **only one** question out of two.

Write your answers on the separate answer paper provided.

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Section A	
1	/11
2	/15
3	/14
4	/13
5	/10
6	/7
7	/10
Total	/80
Section B	
9 OR 10	/20
TOTAL	/100

This question paper consists of 18 printed pages

SECTION A

Answer **all** questions.

Question 1

Figure 1.1 below depicts the molecular structure of a basic unit of collagen.

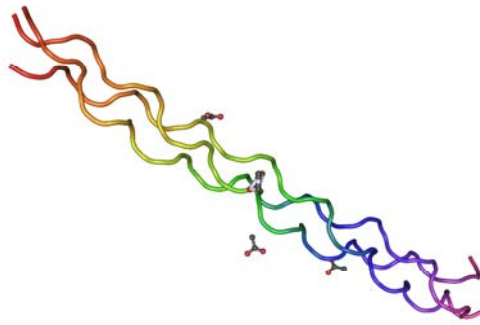


Figure 1.1

(a) State the name given to such a basic unit of collagen. [1]

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(b) Describe how the monomers of this basic unit are joined together to achieve the final molecular configuration as shown in **Figure 1.1**. [3]

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Collagen is normally found in animal connective tissue where its role is as a structural molecule. Figure 1.2 shows an electron micrograph of collagen.

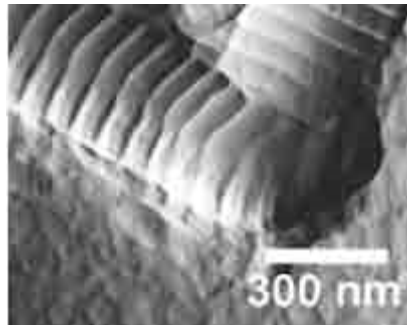


Figure 1.2

(c) Explain the banded appearance of collagen shown in **Figure 1.2**. [2]

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(d) Explain why collagen is able play the role of a structural molecule. [2]

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Another common structural molecule found in nature is cellulose. Cellulose is the main structural molecule in plants.

(e) Compare the structures of cellulose and collagen. [3]

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[Total: 11 marks]

Question 2

The following Figure 2.1 shows an electron micrograph of several cells.

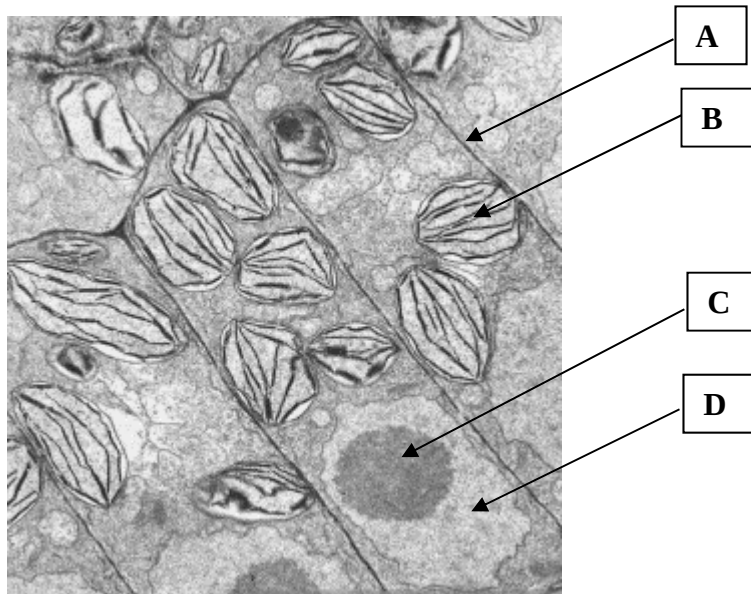


Figure 2.1

(a) Label the organelles **A – D**. [4]

A -

B -

C -

D -

(b) Discuss the role of organelle C. [3]

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(c) In the nucleoplasm of such cells, genetic material can be found. State the nature of this genetic material and briefly describe how this genetic material is organized. [3]

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Such genetic material can take part in processes such as what is shown in **Figure 2.2**.

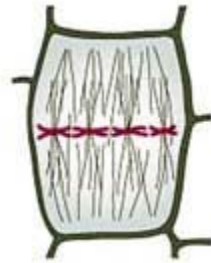


Figure 2.2

(d) With appropriate reasons, precisely identify the process shown in **Figure 2.2**. [3]

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(e) Suggest the significance of the **next stage** of the above process. [2]

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[Total: 15 marks]

Question 3

A set of abundant nucleoid-associated proteins (NAPs) play key functions in organizing the bacterial chromosome and regulating gene transcription globally. Histone-like nucleoid structuring protein (H-NS) is representative of a family of NAPs that are widespread across bacterial species. They have drawn extensive attention due to their crucial function in gene silencing in bacterial pathogens. Figure 3.1 illustrates how H-NS is able to silence genes of bacterial pathogens. (*Information obtained from Biophysical Journal Oct 2015, 109(7)*)

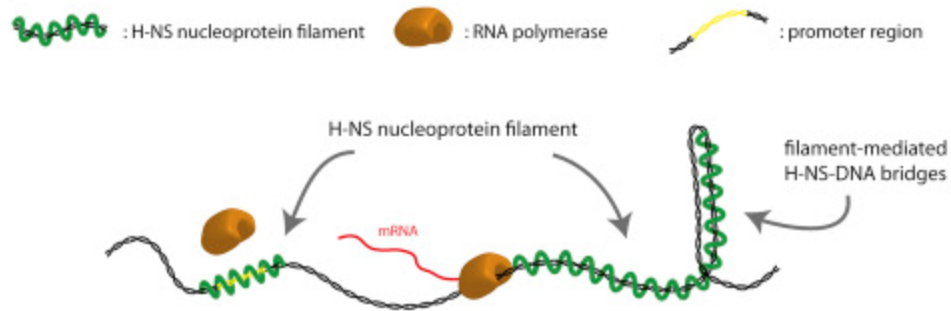


Figure 3.1

(a) State the name of one bacterial pathogen. [1]

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(b) Explain the role of RNA polymerase in a bacterial cell. [3]

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(c) Using the information from **Figure 3.1**, suggest how H-NS is able to silence genes. [2]

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The following figure shows regulation of transcription in the trp operon of a bacterial cell.

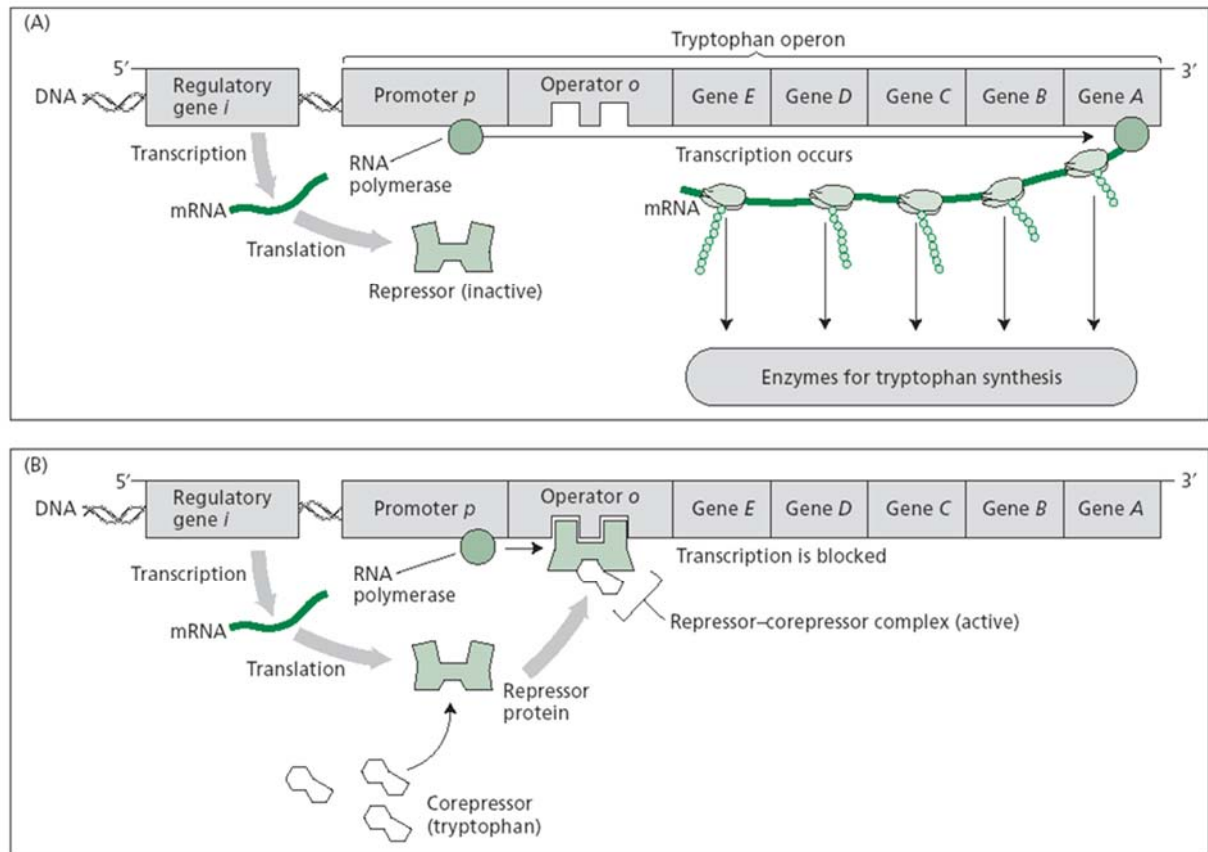


Figure 3.2

(d) Using the information in **figure 3.2**, describe the process of negative regulation in bacterial operons. [4]

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(e) Suggest why tryptophan is the ideal co-repressor for this operon. [2]

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(f) The lac operon is an example of an inducible operon system. Contrast negative regulation of the lac operon with that of the trp operon. [2]

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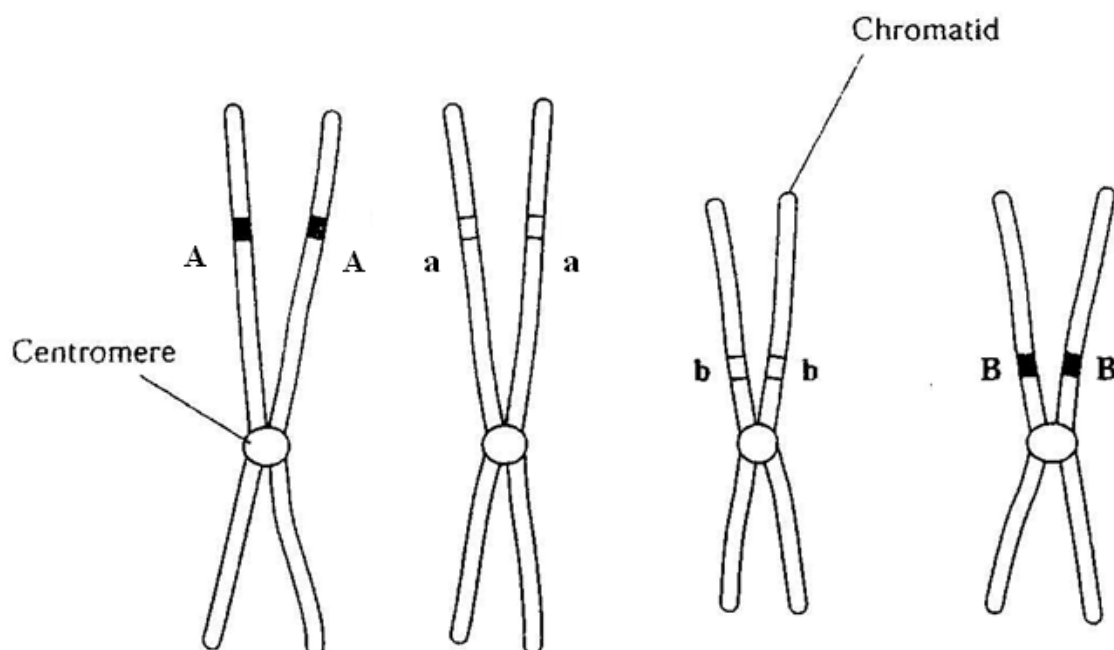
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[Total: 14 marks]

Question 4

The diagram shows the chromosome from a plant cell in which $2n = 4$. The dominant allele **A** of a gene results in red flower and pink spines on the fruit. The recessive allele, **a**, gives yellow flower and green spines. The dominant allele **B** intensifies the colour of the pink spines to red. The plant is heterozygous as shown below:



- (a) The plant was selfed. Give the phenotypic ratio of the offspring. Explain your answer. [3]

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The plant is also heterozygous for another characteristic, the height of the plant. Allele T = tall is dominant to allele t = dwarf. A cross was made between the tall, red-flowered plant with a dwarf, yellow-flowered plant and produced a large number of offspring. **Table 4.1** shows the results.

Table 4.1

Phenotype	Number of offspring
Tall, red-flowered plant	35
Tall, yellow-flowered plant	14
Dwarf, red-flowered plant	16
Dwarf, yellow-flowered plant	33

- (b) The ratio of phenotypes expected in a cross such as this is 1:1:1:1. Chi-squared test was performed on these data giving a calculated value of X^2 of 14.64.

Table 4.2: Distribution of X^2

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (i) Use the calculated value of X^2 and the table of probability provided in **Table 4.2** to find the probability of the results of the cross is due to chance. [1]

Probability

- (ii) State what conclusions may be drawn from the probability found in (b)(i). [2]

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(iii) Explain the difference between the expected and actual results of the test cross. [3]

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(c) Briefly explain how another factor, besides crossing over, can bring about genetic variation in the offspring produced by a sexually reproducing organism (Exclude mutation from your answer). [1]

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In another completely different cross involving humans, the following results were obtained.

Phenotypes	Genotypes	Color of skin
Extremely dark	PPQQRR	6
Very dark	PpQQRR	5
Dark	PpQqRR	4
Intermediate	PpQqRr	3
Light	ppQqRr	2
Very light	ppqqRr	1
Extremely light	ppqqrr	0

(d) What is the term used to describe the range of phenotypes in the above example? [1]

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(e) Explain the genetic basis for the range of phenotypes seen. [2]

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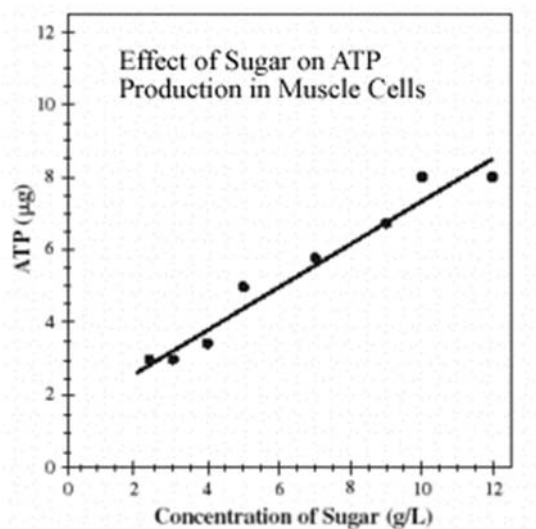
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[Total: 13 marks]

Question 5

The following figure shows the effect of sugar levels on muscle cells.



(a) Describe and explain the effect of sugar on ATP production in muscle cells. [3]

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- (b) Suggest and explain what will happen to ATP production if the concentration of sugar continued to increase. [2]

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The coenzyme cytochrome c oxidoreductase, is the third complex in the electron transport chain, playing a critical role in biochemical generation of ATP (oxidative phosphorylation).

- (c) Explain how the coenzyme cytochrome c oxidoreductase may aid in the production of ATP in the electron transport chain. [3]

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- (d) Actimycin A is known to inhibit the coenzyme cytochrome c oxidoreductase. Predict the effect actimycin A would have on the aerobic respiratory process. [2]

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[Total: 10 marks]

Question 6

The following Figure 6.1 shows the synaptic knob of a synapse found in the sympathetic ganglion of frogs. Such synapses are unique as they contain receptors to the neurotransmitter noradrenaline (NE) on the pre-synaptic membrane itself. These are known as autoreceptors.

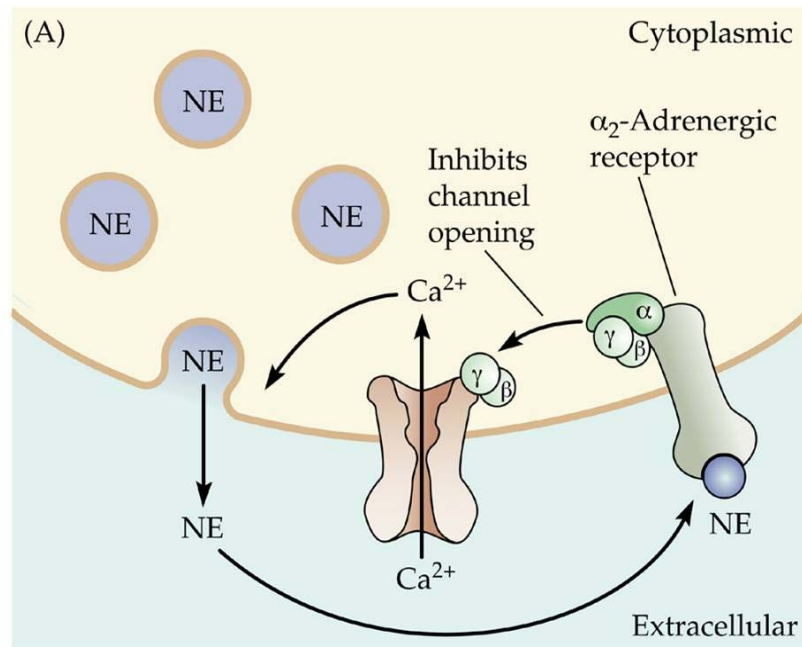


Figure 6.1

(a) State the name given to synapses that use noradrenaline as the neurotransmitter. [1]

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(b) Using the information given in Figure 6.1, explain how the release of noradrenaline at the pre-synaptic membrane may eventually inhibit the synapse itself. [3]

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(c) Suggest the significance of this self-inhibitory effect. [1]

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(d) Describe two functions of synapses. [2]

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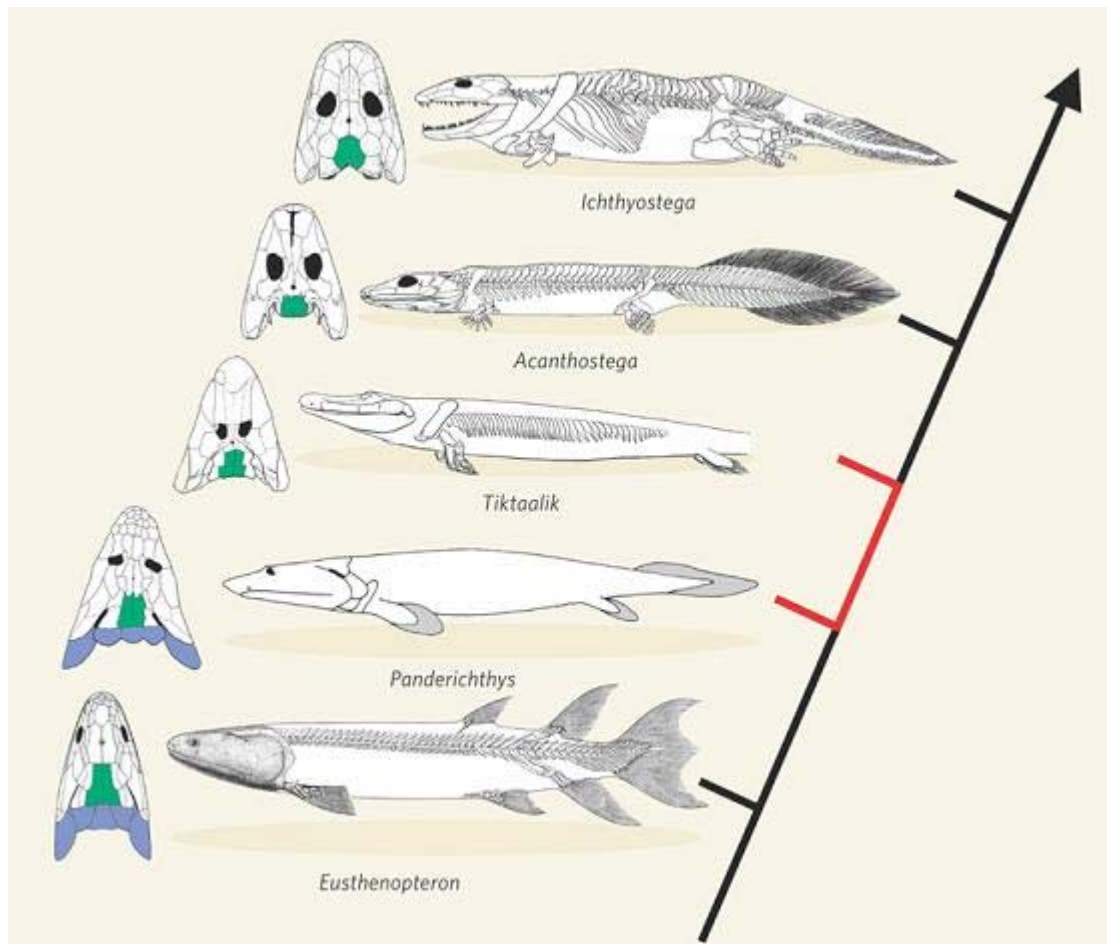
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[Total: 7 marks]

Question 7

Figure 7.1 shows a series of fossils. This series depicts how land-based amphibians could have evolved from fishes. *Tiktaalik* hails from the Late Devonian period, about 360 million years ago, and is both chronologically and morphologically intermediate between two other major fossils in this series, the more fish-like *Panderichthys* and the more tetrapod-like *Acanthostega*.



- (a) State the name given to fossils such as *Tiktaalik* that have characteristics from two seemingly diverse groups of organisms. [1]

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(b) Explain how such fossil records can actually support Darwin's theory of evolution. [4]

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In most respects *Tiktaalik*'s body is fish-like: it has fins and gill arches, just like a fish. However, its skull and especially its limbs mark it as a tetrapod ancestor. Species such as *Panderichthys* had true fins, similar to those of modern ray-finned fishes, consisting of an array of long, thin, spindly bones unsuitable for bearing weight. On the other side of the gap is *Acanthostega*, with true limbs – each containing a radius and an ulna, just like our arms, and outfitted with eight true toes.

(c) State the name given to similar structures such as the limbs of *Acanthostega* and *Tiktaalik*. [1]

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(d) It was believed that the environment *Tiktaalik* evolved in was filled with swampy, silty lagoons. These dirty, unclear water masses also tended to have algae covering its surface. Using this information, describe how the amphibian-like *Acanthostega* could have evolved from the species *Tiktaalik*. [4]

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Stimulus information adapted from: <http://www.patheos.com/blogs/daylightatheism/2006/04/hello-beautiful/> & <https://sciencenotes.wordpress.com/tag/>

[Total: 10 marks]

Section B

Answer **one** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose where appropriate.
Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 8

- (a) Explain the eukaryotic processing of pre-mRNA. [6]
- (b) Compare and contrast between prokaryotic and eukaryotic control of gene expression at the translational and post-translation level. [9]
- (c) Describe the significance of gene amplification. [5]

Question 9

- (a) With reference to the three different stages of cell signaling, describe in detail the sequence of events when a glucagon molecule reaches the liver cell. [10]
- (b) Discuss roles played by proteins in maintaining the potential differences across membranes and in the transmission of nerve impulses along an axon. [7]
- (c) Explain how loss of myelination, which happens in the demyelinating disease multiple sclerosis, disrupts signal transmission in the nervous system. [3]